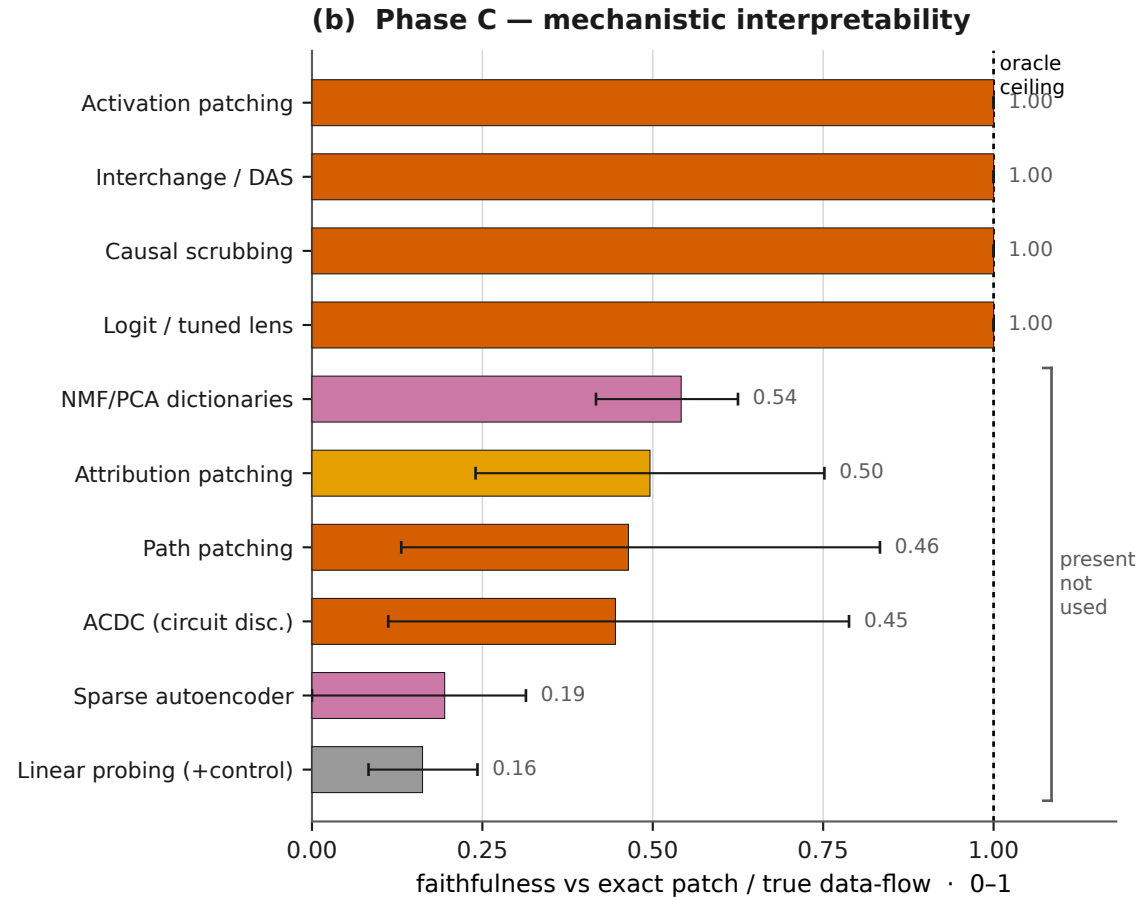
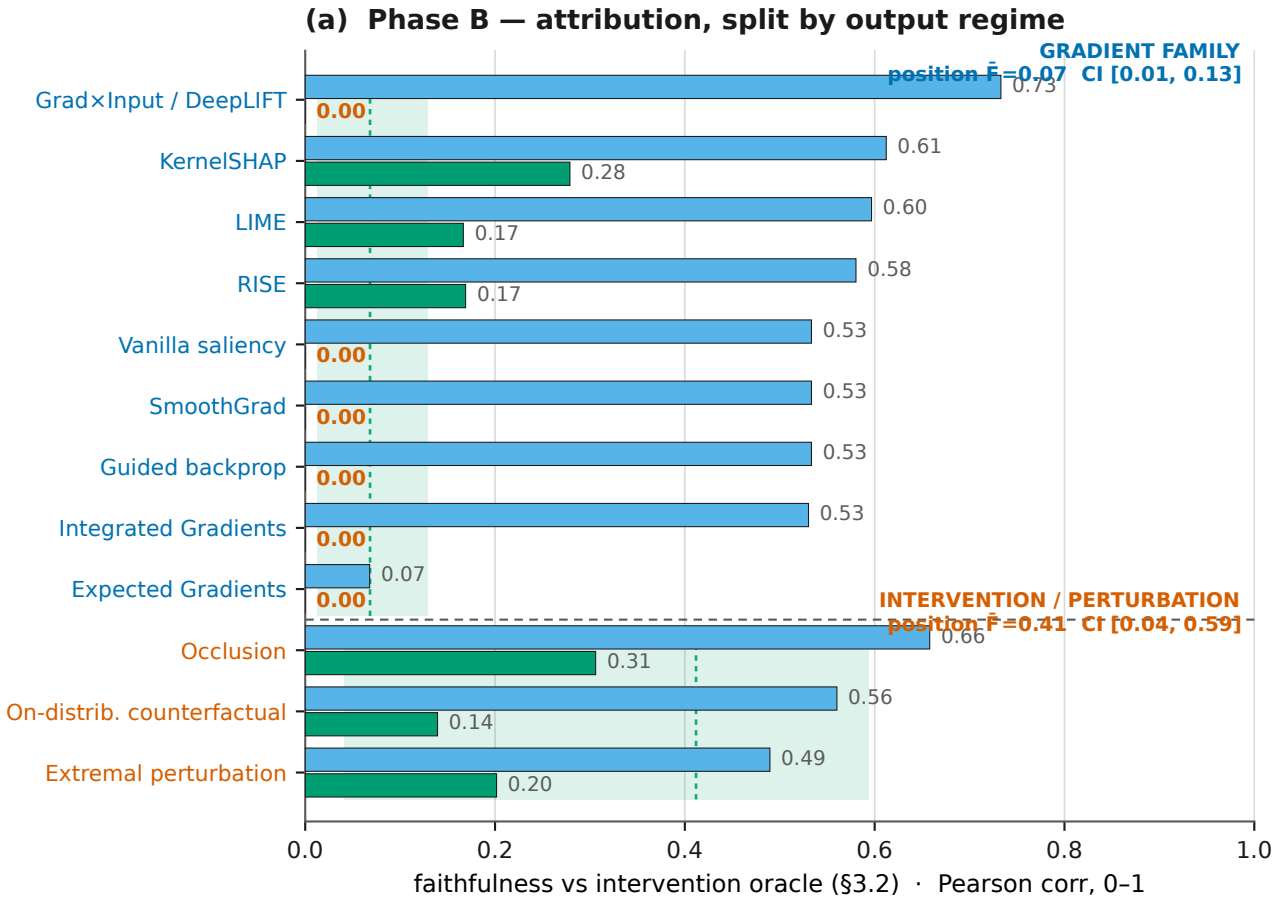


Attribution (Phase B) vs mechanistic interpretability (Phase C): faithfulness on the bit-exact VCS

The 22 Phase-B + Phase-C method rows (a subset of the 30 methods + the oracle = 31-row leaderboard; the cross-tradition scatter is Fig. 2).

On the POSITION/INDEX regime (discrete sprite-position outputs, naive gradient provably zero): the whole gradient family collapses toward chance while intervention methods hold. Causal/intervention bucket 0.412 (95% CI ± 0.390 , n=4) vs gradient/correlational 0.068 (95% CI ± 0.070 , n=9) — a 0.344 faithfulness gap. Activation patching = 1.000 (oracle ceiling) vs vanilla saliency = 0.000 (naive-gradient floor).



(a) output regime
content output position / index output

(b) Phase-C category + uncertainty
causal (exact / partial) gradient approximation dim-reduction / SAE (present-not-used) probing (present-not-used) — 95% CI (bootstrap over)

Data: committed records (leaderboard + bootstrap CIs + faithful-demo); 6 core games, 30-frame horizon; pure read — no experiment re-run. Exact paths & per-number provenance in the Supplement.